

Two-Step Equations in Context

Worksheet 7A.01.04

Name _____

Date _____

Period _____

Frankie solved the equation $6y + 2.5 = 46$. Her work is shown. Use it for questions 1 and 2.

$$6y + 2.5 - 2.5 = 46 - 2.5$$

$$6y = 43.5$$

$$6y \div 6 = 43.5 \div 6$$

$$y = 7.25$$

1. Use substitution to decide whether Frankie's solution is correct.
2. If Frankie is correct, name the properties of equality she used. If she is not, describe her error.

Alishia solved the equation $7x - 4\frac{1}{4} = 44.75$. Her work is shown. Use it for questions 3 and 4.

$$\begin{aligned} 7x - 4\frac{1}{4} - 4\frac{1}{4} &= 44.75 - 4\frac{1}{2} \\ 7x &= 40.25 \\ 7x \div 7 &= 40.25 \div 7 \\ x &= 5.75 \end{aligned}$$

3. Use substitution to decide whether Alishia's solution is correct.
4. Describe the errors Alishia made, then solve the equation correctly.

For questions 5 and 6, solve the equation, then say what the solution means in the situation.

- A gym charges a \$55 joining fee plus a monthly fee. A full year costs \$235. The equation $55 + 12t = 235$ represents the situation.
- Renting a kayak costs \$24 for the first 3 hours and \$4.50 for each additional hour. You spent \$42. The equation $24 + 4.50m = 42$ represents the situation.

For questions 7 through 10, solve the equation and check your solution by substitution.

7. $8 + \frac{z}{4} = 23$

8. $5t - 17.2 = 16.3$

9. $\frac{s}{3} - 0.6 = 1.2$

10. $2.4a + 8 = 27.2$